

5. Project Development Phase

1. INTRODUCTION

The Project Development Phase executes the foundational mechanics of the **ComicCraft** application. This stage moves the design into functional code by realizing the full end-to-end generation sequence: outline generation, story generation, automated panel image rendering, layout synthesis, PDF document exporter, and seamless web routing powered by FastAPI.

2. OBJECTIVES

Phase 5 focuses on building out the operational backend engine and user interaction flow. Key objectives include:

- Develop the core **FastAPI** web application and request handlers.
- Implement automated text generation workflows for story outlines and multi-panel narratives.
- Integrate AI image generation capabilities based on prompt contexts and aesthetic styles.
- Construct dynamic panel layout builders to stitch generated images and narrative text.
- Develop PDF export engines to produce ready-to-read comic book files.
- Establish routing endpoints and integrate frontend interactive Jinja2 templates.

3. PHASE WORKFLOW

The core execution flow follows a continuous data transformation pipeline:

```
FastAPI Route → Outline Generation → Story Generation → Image Generation →  
Layout Building → PDF Export
```

4. DETAILED EXPLANATION

The provided workflow establishes the operational contracts and module interfaces required for production execution. While abstracting full external SDK dependencies, this phase provides a GitHub-ready integration skeleton designed for systematic module coupling. Each stage passes context downstream—from parameter ingestion through AI synthesis to final document compilation.

5. GITHUB-READY CODE / STRUCTURE

Below is the integration skeleton for the primary FastAPI route handling generation requests:

```

from fastapi import APIRouter, Form

router = APIRouter()

@router.post("/generate")
async def generate(
    story_prompt: str = Form(...),
    character_name: str = Form(...),
    setting: str = Form(...),
    tone: str = Form(...),
    art_style: str = Form(...)
):
    # Workflow Pipeline Sequence:
    # generate_outline -> generate_story ->
    # generate_image -> build_comic_layout -> save_pdf

    return {"status": "generation pipeline started"}

```

6. GITHUB FILES & DELIVERABLES

Key Project Repository Artifacts:

- `app/main.py` — FastAPI application entry point and server setup.
- `app/routes.py` — API routing handlers and endpoint definitions.
- `app/gemini_flash.py` — Integration for fast outline generation.
- `app/gemini_pro.py` — Integration for detailed story generation.
- `app/image_generator.py` — AI art and panel rendering integration.
- `app/layout_builder.py` — Panel arrangement and comic layout compiler.
- `app/exporters.py` — PDF rendering and file conversion utilities.
- `templates/` — Jinja2 HTML views for user interactions.

7. EXPECTED OUTCOME

By the conclusion of Phase 5, all core generation, layout, and export services within **ComicCraft** are implemented, integrated, and verified against functional criteria. Execution readiness and feature verification depend on the active repository code and execution setup.

8. PHASE CONCLUSION

Phase 5 provides a structured foundation for the next developmental stage of **ComicCraft**. By connecting core generative engines, layout builders, and API routes within a maintainable GitHub architecture, the application maintains high developer velocity, modularity, and systemic stability.